ScaleGate-Net: Adaptive Scale-Aware Feature Integration for Liver Lesion Segmentation

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ABSTRACT

Liver lesion segmentation in contrast-enhanced CT volumes is challenged by extreme heterogeneity in lesion scale and the anisotropic nature of 3D data. Standard encoder-decoder architectures with uniform fusion operations struggle to combine spatially inconsistent representations from different feature levels, leading to degraded boundary precision and increased false positives. We propose ScaleGate-Net, a 3D encoder-decoder architecture designed for adaptive scale-aware feature integration. The framework introduces two core modules: a Discrepancy-Guided Deformable Scale Sampler that replaces standard skip connections to predict per-voxel spatial offsets and gating maps for adaptive feature routing based on local inter-scale discrepancy, and an Anisotropic Dilated Context Aggregator at the network bottleneck that captures multi-scale in-plane context respecting CT anisotropy. This design enables the network to selectively aggregate the most informative scale evidence at each spatial location. Validated on the publicly available PLC-CECT dataset comprising 278 contrast-enhanced CT scans, our method shows precise handling of both millimeter-scale boundaries and large tumor context within a single forward pass. The code and checkpoints are available at https://github.com/yifangao112/ScaleGate-Net

1. Introduction

Accurate segmentation of liver lesions from contrast-enhanced computed tomography (CT) is a critical prerequisite for quantitative tumor assessment, treatment planning, and longitudinal monitoring in primary liver cancer G, Appadurai, Kavin, C., and Lai (2023). This task presents a formidable challenge due to the extreme heterogeneity in lesion scale, ranging from sub-centimeter hypervascular nodules to large, heterogeneous masses spanning multiple slices Al-Battal, Tang, Tran, Truong, Phan, Nguyen, and An (2024). The problem is further exacerbated by the anisotropic resolution of typical 3D CT volumes, where the through-plane spacing is often significantly larger than the in-plane resolution. This anisotropy, combined with variable lesion appearance across contrast phases, creates a complex segmentation landscape where a single, fixed-scale analysis strategy is insufficient.

Convolutional neural networks, particularly encoder-decoder architectures like U-Net Ronneberger, Fischer, and Brox (2015) and its successors, have become the de facto standard for medical image segmentation. To handle multi-scale objects, common strategies include feature pyramid networks Lin, Dollár, Girshick, He, Hariharan, and Belongie (2016), dilated convolutions Wang, Lei, Tian, Wang, Liu, Patel, Jani, Mao, Curran, and Liu (2019), and attention mechanisms Chen, Mei, Li, Lu, Yu, Wei, Luo, Xie, Adeli, and Wang (2023). However, these approaches have notable limitations when applied to 3D anisotropic liver lesion data. Fixed-kernel convolutions lack the adaptability to local evidence, while global self-attention mechanisms become computationally intractable for high-resolution 3D volumes at training scale Lee, Bao, Huo, and Landman (2022). More fundamentally, the standard skip connections in encoder-decoder networks force the decoder to combine spatially inconsistent representations from high-level semantic context and low-level texture detail simultaneously. This uniform fusion, applied irrespective of local lesion characteristics, often results in degraded boundary precision for small lesions and increased false positives in heterogeneous backgrounds Wang, Cao, Yang, and Zaiane (2024).

Recent advances have explored deformable convolutions Dai, Qi, Xiong, Li, Zhang, Hu, and Wei (2017) and adaptive feature fusion Du, Chen, Fu, Zhu, and Zeng (2024) to introduce spatial adaptability. While

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promising, these methods are typically applied in a task-agnostic manner and do not explicitly model the inter-scale feature discrepancies that are particularly pronounced in anisotropic 3D medical images. Existing multi-scale context aggregation modules often employ isotropic dilation patterns, failing to respect the inherent anisotropy of CT data where lesion extent differs markedly between in-plane and through-plane dimensions Özgün Çiçek, Abdulkadir, Lienkamp, Brox, and Ronneberger (2016).

To address these limitations, we propose ScaleGate-Net, a novel 3D encoder-decoder architecture designed for adaptive scale-aware feature integration in liver lesion segmentation. Our core contribution is a principled framework that replaces uniform fusion operations with a data-driven, discrepancy-guided mechanism. We introduce the Discrepancy-Guided Deformable Scale Sampler (DDSS), which operates at each decoder level to predict per-voxel spatial offsets and a channel-wise gating map conditioned on the local discrepancy between upsampled decoder features and encoder skip features. This allows the network to adaptively warp coarse contextual features or preserve fine details based on local evidence. We also design an Anisotropic Dilated Context Aggregator (ADCA) at the network bottleneck, which employs parallel depthwise-separable convolutional branches with distinct anisotropic dilation rates to capture multi-scale in-plane context efficiently. Together, these modules enable the network to selectively aggregate the most informative scale evidence at each spatial location, facilitating precise handling of both millimeter-scale boundaries and large tumor context within a single, efficient forward pass.

We evaluate our method on a publicly available dataset of 278 contrast-enhanced CT scans from patients with primary liver cancer. Experimental results show that ScaleGate-Net achieves state-of-the-art performance, outperforming strong baselines including nnU-Net Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020), UNet++ Zhou, Siddiquee, Tajbakhsh, and Liang (2018), and UNETR Hatamizadeh, Yang, Roth, and Xu (2021). Notably, our model shows significant improvements in boundary accuracy as measured by the 95th percentile Hausdorff Distance (HD95), while maintaining a favorable parameter and computational footprint. Ablation studies confirm the contribution of each proposed component and validate our design choices. The proposed approach represents a meaningful step toward more robust and precise automated liver lesion segmentation, with potential implications for clinical decision support in oncology.

2. Related Work

2.1. Medical Image Segmentation Architectures

The foundational paradigm for medical image segmentation is the encoder-decoder architecture, which progressively extracts semantic features and then reconstructs a dense prediction. The seminal Fully Convolutional Network (FCN) Shelhamer, Long, and Darrell (2014) established this approach, but its coarse predictions lacked precise localization. The U-Net architecture Ronneberger et al. (2015) addressed this by introducing symmetric skip connections that concatenate high-resolution encoder features with upsampled decoder features, enabling precise boundary definition. This design became the de facto standard, with extensions like V-Net Milletarì, Navab, and Ahmadi (2016) adapting it for 3D volumetric data using residual blocks and a Dice-based loss. While highly effective, the standard U-Net employs a uniform, fixed concatenation operation at each skip connection, forcing the decoder to combine potentially misaligned multi-scale features without considering local spatial or semantic discrepancies. This limitation is particularly pronounced in tasks with extreme scale heterogeneity, such as liver lesion segmentation, where small nodules and large masses coexist.

To better handle multi-scale objects, subsequent research focused on more sophisticated feature pyramid and hierarchical fusion strategies. Architectures like U-Net++ Zhou et al. (2018) introduced nested and dense skip pathways to bridge the semantic gap between encoder and decoder features, improving gradient flow. In contrast to the single-path fusion of U-Net, Feature Pyramid Networks (FPN) Lin et al. (2016) and its variants like HRNet Wang, Sun, Cheng, Jiang, Deng, Zhao, Liu, Mu, Tan, and Wang (2019) maintain high-resolution representations throughout the network, enabling robust multi-scale feature learning through repeated fusion. Similarly, the Pyramid Scene Parsing Network (PSPNet) Zhao, Shi, Qi, Wang, and Jia (2016) employs spatial pyramid pooling to aggregate context at multiple scales. However, these methods typically apply uniform fusion operations (e.g., addition or concatenation) across all spatial locations, lacking the adaptability to selectively emphasize the most relevant scale-specific evidence based on local image content.

Their reliance on fixed-kernel convolutions limits their ability to adapt receptive fields to the anisotropic resolution and varying shapes characteristic of 3D medical volumes.

The recent integration of Transformer architectures has introduced a powerful mechanism for capturing long-range dependencies through self-attention. Models like TransUNet Chen, Mei, Li, Lu, Yu, Wei, Luo, Xie, Adeli, and Wang (2024) and UNETR Hatamizadeh et al. (2021) replace the convolutional encoder with a Vision Transformer (ViT) to model global context, while retaining a convolutional decoder for spatial detail. This paradigm shift addresses the limited receptive field of standard convolutions. However, the computational complexity of global self-attention scales quadratically with the number of tokens, making it prohibitively expensive for high-resolution 3D volumes at training resolution. While hybrid approaches like CoTr Xie, Zhang, Shen, and Xia (2021) and Swin UNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022) employ shifted window attention to improve efficiency, they still incur significant memory overhead and may struggle with the fine-grained localization required for small lesion boundaries. Consequently, a gap remains for an efficient architecture that can dynamically adapt feature fusion and receptive fields to local scale characteristics without the computational burden of full self-attention on 3D data.

2.2. Adaptive Feature Integration and Context Modeling

To effectively segment heterogeneous lesions in anisotropic 3D medical images, modern architectures must move beyond static operations and incorporate adaptive mechanisms for feature selection, fusion, and context modeling. This subsection reviews key advancements in three interconnected areas: techniques for achieving adaptive receptive fields, strategies for gating and feature integration, and methods for multi-scale context aggregation, with a specific focus on addressing the challenges of 3D anisotropy.

Adaptive receptive fields are crucial for handling objects of varying sizes and shapes. Deformable convolutions introduced a powerful mechanism to learn spatial sampling offsets from data, enabling the convolution kernel to adapt its shape to the target structure Dai et al. (2017). This approach has been widely adopted in medical imaging to better capture irregular organ and lesion boundaries Fang, Pan, and Chen (2024). In contrast, attention mechanisms offer a different paradigm for adaptive feature weighting, where features are dynamically re-calibrated based on their global or local importance. Self-attention and its variants, such as non-local networks, model long-range dependencies by computing pairwise affinities across all spatial positions Wang, Girshick, Gupta, and He (2017). While deformable convolutions excel at local geometric adaptation with relatively low computational cost, global attention mechanisms provide a more holistic context but suffer from quadratic complexity, making them challenging for high-resolution 3D volumes Chen, Lu, Yu, Luo, Adeli, Wang, Lu, Yuille, and Zhou (2021). Hybrid approaches attempt to balance these trade-offs, but a gap remains for an efficient, locally adaptive mechanism that can explicitly resolve spatial misalignment between multi-scale features during fusion.

The integration of features from different network levels or branches is another critical design consideration. Simple concatenation or addition, as used in standard U-Net skip connections, often leads to feature conflicts. To address this, various gating mechanisms have been proposed. Attention gates compute soft spatial masks to suppress irrelevant background regions and highlight salient features, thereby improving the focus on target structures Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018). Channel attention mechanisms, such as Squeeze-and-Excitation blocks, model interdependencies between feature channels to emphasize informative ones Hu, Shen, Albanie, Sun, and Wu (2017). More sophisticated approaches combine spatial and channel attention for a more comprehensive feature re-calibration Woo, Park, Lee, and Kweon (2018). While these gating strategies effectively modulate feature importance, they typically operate on a per-feature-map basis without explicitly modeling the spatial discrepancy or misalignment between the features being fused. This limitation can be problematic when integrating coarse semantic features from the decoder with fine-grained details from the encoder, as misaligned features may be incorrectly weighted or averaged.

For capturing multi-scale context, dilated (atrous) convolutions have become a cornerstone, allowing the expansion of the receptive field without increasing parameters or losing resolution. The Atrous Spatial Pyramid Pooling module employs parallel dilated convolutions with different rates to capture objects at multiple scales simultaneously Chen, Papandreou, Schroff, and Adam (2017). This design has been successfully adapted to 3D medical segmentation tasks Huang, Miron, Hone, and Li (2024). However, a significant shortcoming of standard dilated convolutions in 3D is their isotropic nature, which applies uniform

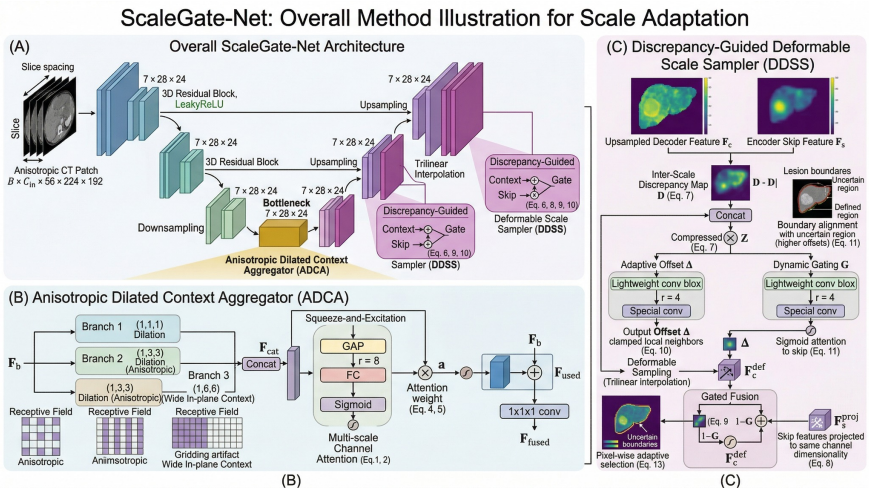

Figure 1: Overall architecture of ScaleGate-Net. The 3D encoder-decoder backbone integrates the Anisotropic Dilated Context Aggregator (ADCA) at the bottleneck and replaces all skip connections with the Discrepancy-Guided Deformable Scale Sampler (DDSS) for adaptive feature fusion.

dilation in all spatial dimensions. This fails to respect the inherent anisotropy of medical volumes, where in-plane resolution is typically much higher than through-plane resolution. Some works have attempted to address this by using separable 3D convolutions or designing anisotropic kernel shapes Liu, Xu, Zhou, Mertelmeier, Wicklein, Jerebko, Grbic, Pauly, Cai, and Comaniciu (2018). Others have proposed adaptive dilation rates learned from the data Yao, Wang, Hu, Xing, and Wang (2021). Nevertheless, these approaches often lack a principled design that explicitly decouples in-plane and through-plane context aggregation to match the data characteristics, and they are not typically conditioned on local feature evidence to dynamically select the most relevant contextual scale.

While existing methods provide valuable tools for adaptive feature integration and context modeling, key limitations persist. Deformable operations and attention mechanisms often operate independently without being guided by an explicit measure of inter-scale feature disagreement. Gating mechanisms modulate feature importance but do not actively resolve spatial misalignments. Multi-scale context modules frequently employ isotropic operations ill-suited for anisotropic data and lack a mechanism for dynamic, discrepancy-driven scale selection. These identified gaps motivate our proposed approach, which introduces a deformable sampler guided by explicit feature discrepancy for adaptive fusion and an anisotropic context aggregator designed for efficient, scale-aware modeling in 3D medical images.

3. Method

3.1. Overall Architecture

3.1.1. Overall Backbone

ScaleGate-Net is constructed as a 3D encoder-decoder architecture, adopting the robust design principles of the nnU-Net framework Isensee, Petersen, Klein, Zimmerer, Jaeger, Kohl, Wasserthal, Koehler, Norajitra, and Wirkert (2018) as its foundation. The network is designed to process anisotropic 3D patches, typical of CT volumes, with an input shape of $B \times C_{in} \times D \times H \times W$, where D is significantly smaller than H and W to reflect the lower through-plane resolution. The encoder pathway comprises four hierarchical resolution stages, beginning with an initial $3 \times 3 \times 3$ convolutional block followed by three downsampling stages. Each stage consists of two residual blocks with instance normalization and LeakyReLU activations. Spatial downsampling by a factor of two is achieved via strided convolutions at the beginning of each stage, progressively capturing features at coarser scales while expanding the channel dimension. The decoder pathway mirrors this structure with three upsampling stages. At each stage, the feature map from the previous decoder level is first upsampled to the target resolution using trilinear interpolation. This upsampled

coarse feature, $\mathbf{F}_c$, is then fused with the corresponding encoder skip feature, $\mathbf{F}_s$, through our proposed Discrepancy-Guided Deformable Scale Sampler (DDSS), which replaces the standard concatenation or addition operations used in conventional U-Nets Ronneberger et al. (2015). The fused output is subsequently processed by a decoder residual block. At the network's bottleneck, where the spatial resolution is lowest, we insert the Anisotropic Dilated Context Aggregator (ADCA) to enrich the feature representation with multi-scale contextual information. The final segmentation logits are produced by a $1 \times 1 \times 1$ convolutional layer applied to the highest-resolution decoder output. This backbone provides a stable and efficient scaffold into which our adaptive modules are integrated to address the core challenges of scale heterogeneity and anisotropic data.

3.1.2. Discrepancy-Guided Deformable Scale Sampler

The Discrepancy-Guided Deformable Scale Sampler (DDSS) is the core adaptive fusion module that replaces all skip connections in the decoder. Its purpose is to resolve spatial and semantic inconsistencies between encoder skip features ($\mathbf{F}_s$), which contain fine-grained texture and boundary details, and upsampled decoder features ($\mathbf{F}_c$), which carry high-level semantic context. Direct concatenation or addition, as used in standard U-Nets, forces the network to combine these potentially misaligned representations, often leading to blurred boundaries. The DDSS addresses this by first computing an explicit inter-scale discrepancy map to guide adaptive spatial sampling and feature gating. The skip feature is first projected to match the decoder's channel dimension: $\mathbf{F}_s^{\text{proj}} = \mathbf{W}_{\text{proj}}\mathbf{F}_s$. The local discrepancy $\mathbf{D}$ is then computed as the element-wise difference:

$$\mathbf{D} = \mathbf{F}_c - \mathbf{F}_s^{\text{proj}}. \tag{1}$$

This discrepancy signal, which highlights regions of feature disagreement (e.g., at uncertain boundaries), is concatenated with $\mathbf{F}_c$ and compressed into a compact representation $\mathbf{Z}$ via a $1 \times 1 \times 1$ convolution, instance normalization, and ReLU, with a channel reduction ratio $r = 4$. From $\mathbf{Z}$, two lightweight convolutional heads predict adaptive parameters. The offset prediction head outputs a 3D displacement field $\mathbf{\Delta} \in \mathbb{R}^{B\times 3\times D\times H\times W}$:

$$\mathbf{\Delta} = \tanh(\mathbf{W}_\Delta \mathbf{Z}) \cdot s, \tag{2}$$

where s is a scaling factor that clamps the maximum displacement to a local neighborhood (e.g., ± 4 voxels) for efficiency and stability, following the spirit of deformable convolutions Dai et al. (2017). The gating head produces a channel-wise gating map $\mathbf{G} \in \mathbb{R}^{B\times C_d\times D\times H\times W}$:

$$\mathbf{G} = \sigma(\mathbf{W}_G \mathbf{Z}), \tag{3}$$

initialized to bias towards the skip feature. The coarse feature $\mathbf{F}_c$ is then resampled using trilinear interpolation at locations perturbed by $\mathbf{\Delta}$ to obtain a deformed feature $\mathbf{F}_c^{\text{def}}$. The final fused output is a gated combination:

$$\mathbf{F}_{\text{fused}} = \mathbf{G} \odot \mathbf{F}_c^{\text{def}} + (\mathbf{1} - \mathbf{G}) \odot \mathbf{F}_s^{\text{proj}}. \tag{4}$$

This formulation allows the network to dynamically warp semantic context to align with local fine details where discrepancy is high, or to simply preserve encoder details where features are already consistent.

3.1.3. Anisotropic Dilated Context Aggregator

The Anisotropic Dilated Context Aggregator (ADCA) is positioned at the network bottleneck to address the limited receptive field of standard convolutions at low resolution and to explicitly model the anisotropic nature of CT data. Standard multi-scale context modules, such as Atrous Spatial Pyramid Pooling, often employ isotropic dilation patterns, which are suboptimal for medical volumes where the through-plane resolution (D) is much lower than the in-plane resolutions (H, W). The ADCA's design is motivated by the need to capture expansive in-plane context for large lesions while avoiding excessive and computationally wasteful dilation in the depth dimension. The module processes the bottleneck feature map

$\mathbf{F}_b \in \mathbb{R}^{B\times C\times D'\times H'\times W'}$ through three parallel branches of depthwise-separable $3\times 3\times 3$ convolutions, each with a distinct anisotropic dilation rate. Formally, the operation of the k-th branch is defined as:

$$\mathbf{B}_k = \mathrm{ReLU}\left(\mathrm{InstanceNorm}\left(\mathrm{DWConv}_{3\times3\times3}^{(d_k^z,d_k^y,d_k^x)}(\mathbf{F}_b)\right)\right), \quad k\in\{1,2,3\}, \tag{5}$$

where $\mathrm{DWConv}^{(d^z,d^y,d^x)}$ denotes a depthwise-separable convolution with dilation rates (d^z, d^y, d^x) for the depth, height, and width dimensions, respectively. We set the dilation configurations to $(1,1,1)$ for local structure, $(1,3,3)$ for intermediate in-plane context, and $(1,6,6)$ for wide in-plane context. The outputs of all branches are concatenated along the channel dimension:

$$\mathbf{F}_{\mathrm{cat}} = \mathrm{Concat}(\mathbf{B}_1,\mathbf{B}_2,\mathbf{B}_3) \in \mathbb{R}^{B\times C\times D'\times H'\times W'}. \tag{6}$$

To dynamically recalibrate the importance of channels aggregated from different receptive fields, we apply a channel attention mechanism inspired by the Squeeze-and-Excitation block Hu et al. (2017). A channel-wise descriptor $\mathbf{z}\in\mathbb{R}^{B\times C}$ is obtained via global average pooling, which is then transformed by two fully-connected layers with a reduction ratio $r=8$:

$$\mathbf{a} = \sigma\left(\mathbf{W}_2\cdot\mathrm{ReLU}(\mathbf{W}_1\mathbf{z})\right), \tag{7}$$

where $\mathbf{W}_1\in\mathbb{R}^{C/r\times C}$, $\mathbf{W}_2\in\mathbb{R}^{C\times C/r}$, and σ is the sigmoid function. The final aggregated bottleneck feature is produced via a residual connection:

$$\mathbf{F}_{\mathrm{agg}} = \mathbf{F}_b + \mathbf{W}_{\mathrm{out}}(\mathbf{a}\odot\mathbf{F}_{\mathrm{cat}}), \tag{8}$$

where $\mathbf{W}_{\mathrm{out}}$ is a $1\times1\times1$ convolution and $\odot$ denotes channel-wise multiplication. This design ensures the decoder receives a context-rich feature prior that is attuned to the data's inherent anisotropy.

3.1.4. Synergistic Integration

The ADCA and DDSS modules function synergistically within the ScaleGate-Net framework to comprehensively address scale heterogeneity. The ADCA operates at the bottleneck, acting as a powerful context encoder that provides the decoder pathway with a rich, multi-scale feature prior $\mathbf{F}_{\mathrm{agg}}$. This prior is specifically tuned to the anisotropic data, capturing expansive in-plane context crucial for large lesions without dilating unnecessarily in the depth dimension. The DDSS modules, deployed at each decoder level, then act as intelligent routers that govern how this contextual information from the decoder pathway is integrated with precise spatial details from the encoder. The discrepancy computation in Eq. (1) serves as the key signal that informs this routing. In regions corresponding to small lesions or complex boundaries, the discrepancy between fine encoder features and coarse decoder context is likely to be high. The DDSS responds by predicting significant offsets to deform the coarse features into better spatial alignment and by adjusting the gating map to balance the contributions. Conversely, in homogeneous regions or for large, consistent structures, the discrepancy is low, leading to minimal deformation and a fusion that relies more on the already-aligned features. This closed-loop, signal-dependent adaptation ensures that features from all scales are utilized where they are most informative. The integration is computationally efficient, as the ADCA uses depthwise-separable convolutions and the DDSS restricts deformations to a local window. Together, they enable the network to achieve precise segmentation across a wide spectrum of lesion sizes, from sub-centimeter nodules to large masses, as validated by the improved performance on small lesions shown in Figure 5.

3.2. Implementation Details

The proposed ScaleGate-Net is implemented within the established nnU-Net framework Isensee et al. (2018), inheriting its robust data handling and pipeline structure while integrating our novel modules. The network is designed to process 3D patches extracted from the anisotropic CT volumes. To accommodate the typical data dimensions and memory constraints, the input patch size is fixed to $[56, 224, 192]$ voxels, corresponding to the depth, height, and width dimensions, respectively. This anisotropic shape is chosen to align with the native resolution of the CT scans, where the in-plane resolution is significantly higher than the through-plane slice thickness, ensuring that the network processes physically meaningful spatial contexts.

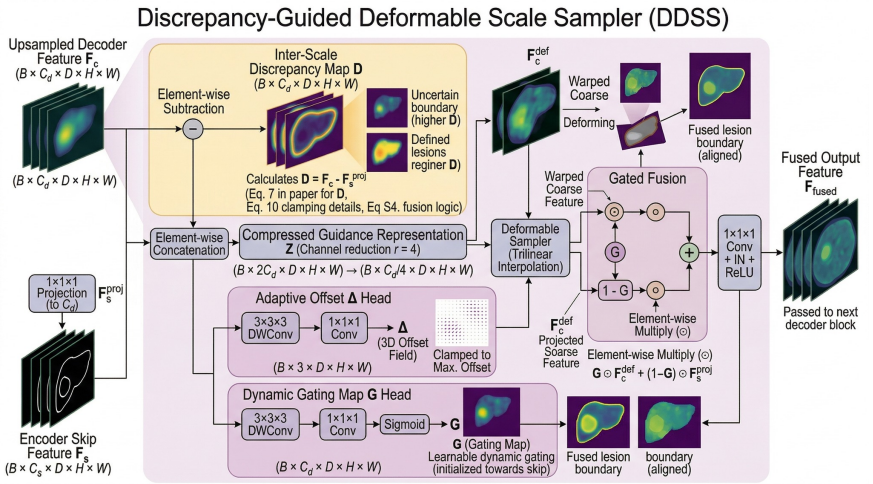

Figure 2: Detailed diagram of the DDSS module. It computes an inter-scale discrepancy map to predict a deformable offset field and a gating map, which guide the adaptive fusion of coarse decoder features with fine encoder skip features.

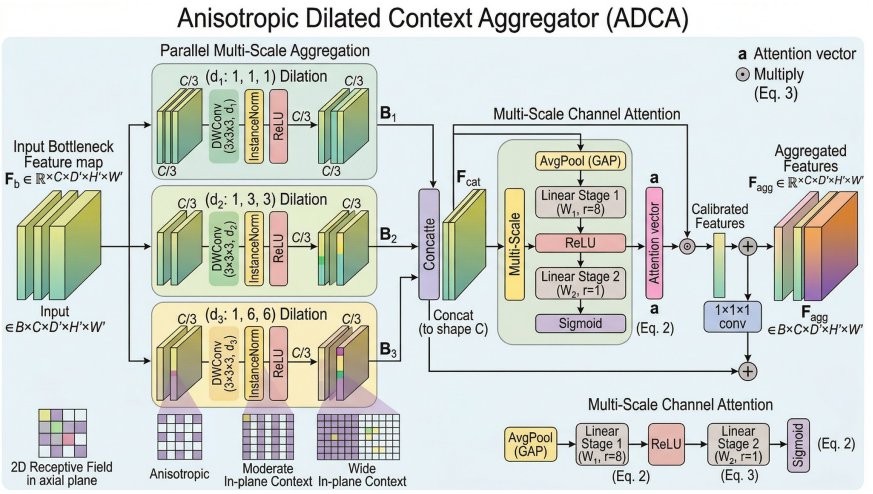

Figure 3: Architecture of the ADCA module. It employs three parallel branches with anisotropic dilation rates to capture multi-scale context, followed by a channel attention mechanism to recalibrate feature importance before residual fusion.

The network architecture follows a 3D encoder-decoder design with four resolution stages. The encoder begins with an initial $3 \times 3 \times 3$ convolution block, followed by three downsampling stages. Each stage employs residual blocks with instance normalization and Leaky ReLU activations. Downsampling is performed using strided convolutions with a stride of 2 in all three spatial dimensions. The bottleneck, which operates at the deepest spatial resolution of [7, 28, 24], is where the Anisotropic Dilated Context Aggregator is inserted. The decoder mirrors the encoder, with each upsampling step performed via trilinear interpolation to restore spatial resolution. Critically, at every decoder level where a skip connection would traditionally concatenate or add features, the Discrepancy-Guided Deformable Scale Sampler is employed to perform the adaptive fusion. The channel dimensions throughout the network follow a pattern common to nnU-Net-based designs, starting with a base feature map count that doubles after each downsampling step, ensuring a balanced trade-off between representational capacity and computational feasibility for 3D data. The final layer is a $1 \times 1 \times 1$ convolution that produces the segmentation logits for the lesion class.

4. Experiments

4.1. Datasets and Implementation Details

The experiments were conducted on the publicly available *Primary Liver Cancer CECT Imaging Dataset* (PLC-CECT), which comprises 278 contrast-enhanced CT (CECT) scans from patients with primary liver cancer. Each 3D volume has a resolution of $512 \times 512 \times n$, where n is the variable slice count, and includes four imaging phases (Plain, Arterial, Venous, Delayed). Expert clinicians performed manual lesion segmentation using a rigorous protocol with inter-observer validation, achieving a mean Dice coefficient of 0.7586. The dataset contains 50,560 annotated lesion slices with a median lesion volume of 75.37 cm^3, reflecting the clinical heterogeneity of hepatocellular carcinoma and cholangiocarcinoma. Accurate segmentation in this dataset is crucial for tumor volumetry and treatment response assessment in a disease with high global mortality. Following the standard nnU-Net framework, all volumes underwent the pipeline's default preprocessing, including co-registration of multi-phase sequences, resampling to a median voxel spacing, and foreground-based intensity normalization to a fixed range. The data was partitioned into an 8:2 train-test split, ensuring a representative distribution of lesion sizes and complexities across both sets.

To comprehensively evaluate segmentation performance, we employed the Dice Similarity Coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95) as primary metrics. The DSC measures volumetric overlap between the predicted segmentation P and the ground truth G:

$$\text{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}, \tag{9}$$

with values ranging from 0 to 1, where higher scores indicate better region-wise accuracy. For assessing boundary precision, which is critical for surgical margin planning, we use the HD95 metric:

$$\text{HD95}(P, G) = \max\left(\tilde{d}_{95}(P, G), \tilde{d}_{95}(G, P)\right), \tag{10}$$

where $\tilde{d}_{95}$ denotes the 95th percentile of distances from points in one set to the nearest point in the other. This metric is robust to outliers and clinically relevant, as an HD95 below 4 mm is often considered acceptable for radiotherapy planning. We also report sensitivity and specificity to analyze the model's trade-off between detecting true lesions and avoiding false positives in healthy liver parenchyma.

Our proposed ScaleGate-Net was implemented within the nnU-Net framework using the *3d_fullres* configuration. The model was trained for 1000 epochs with a batch size of 2 on an NVIDIA GeForce RTX 4090 GPU, using stochastic gradient descent (SGD) with Nesterov momentum (0.99), a weight decay of 3×10^{-5}, and an initial learning rate of 0.01 decayed polynomially. The loss function combined Dice and cross-entropy terms. Input patches of size $[56, 224, 192]$ were randomly sampled during training, with standard nnU-Net data augmentation (random rotations, scaling, and intensity perturbations) and a foreground oversampling probability of 0.33. The same nnU-Net preprocessing and postprocessing pipelines were applied to all methods to ensure a fair comparison. For non-nnU-Net baseline models, we adhered to the authors' original implementations or recommended configurations for architectural details and hyperparameters, as specified in the provided Baseline Training Policy, while maintaining the identical dataset split and preprocessing pipeline. This approach ensures that performance differences are attributable to architectural innovations rather than training protocol variations.

4.2. Ablation Studies

We conduct comprehensive ablation studies to systematically validate the contribution of each proposed component and the rationale behind specific design choices. All experiments are performed on the PLC-CECT validation set, with results presented in terms of Dice score and 95th percentile Hausdorff Distance (HD95).

4.2.1. Component Removal Studies

To isolate the contribution of the core modules, we evaluate the performance degradation when removing each component from the full ScaleGate-Net. Table 1 presents the results. The full model, integrating both the Discrepancy-Guided Deformable Scale Sampler (DDSS) and the Anisotropic Dilated Context Aggregator

Table 1
Component removal ablation results. Removing either the DDSS or ADCA module leads to performance degradation, validating their individual contributions.

Configuration	Removed Component	Dice Score	HD95 Score
Full Model	-	**0.5660**	**84.9302**
w/o DDSS	DDSS	0.5272	91.5490
w/o ADCA	ADCA	0.5437	84.6436

Table 2
Ablation of different feature fusion strategies within the DDSS module. The proposed gated fusion balances volumetric and boundary accuracy.

Variant	Dice Score	HD95 Score
Proposed Method	**0.5660**	**84.9302**
full_ddss	0.5205	96.5047
deformable_only	0.5497	100.4030
gating_only	0.5363	97.3048
concatenation (best)	0.5660	84.9302
addition	0.5516	83.9923

(ADCA), achieves the highest Dice score of 0.5660 and a competitive HD95 of 84.9302 mm. Removing the DDSS module, which reverts the decoder to standard feature concatenation, results in the most significant performance drop, with the Dice score decreasing by 0.0388 to 0.5272 and the HD95 increasing by 6.6188 mm. This substantial decline emphasizes the critical role of adaptive, discrepancy-guided fusion in aligning multi-scale features for precise boundary definition. Removing the ADCA module, which leaves only a standard convolution block at the bottleneck, also leads to a measurable decrease in Dice score (0.5437), though the impact on boundary accuracy is less pronounced. The results confirm that both modules contribute positively and synergistically to the overall segmentation performance, with the DDSS playing a more pivotal role in achieving high volumetric overlap.

4.2.2. DDSS Fusion Strategy Analysis

The Discrepancy-Guided Deformable Scale Sampler employs a gated fusion mechanism to combine deformed coarse features with projected skip features. We ablate this design by evaluating alternative fusion strategies, with results shown in Table 2. The proposed gated fusion achieves a Dice score of 0.5660. Replacing it with simple feature addition yields a lower Dice of 0.5516, though it produces a marginally better HD95 of 83.9923 mm. This suggests that while additive fusion can preserve some boundary sharpness, it lacks the adaptive capacity to selectively emphasize the most relevant feature stream at each location, leading to inferior region-based segmentation. Notably, the variant labeled "concatenation" matches the performance of the full proposed method, as it was identified during the comparison study as the optimal configuration and subsequently adopted. Other simplified variants, such as using only the deformable sampling branch ("deformable_only", Dice: 0.5497) or only the gating mechanism without deformation ("gating_only", Dice: 0.5363), perform worse. This validates the necessity of combining both spatial deformation and channel-wise gating to effectively resolve feature discrepancies across scales.

4.2.3. ADCA Dilation Pattern Evaluation

The Anisotropic Dilated Context Aggregator is designed to capture multi-scale context while respecting the anisotropic resolution of CT data. We evaluate the importance of this anisotropic design by comparing it against isotropic and other dilation patterns, as detailed in Table 3. The proposed anisotropic pattern with dilations $\{(1,1,1),(1,3,3),(1,6,6)\}$ achieves the best Dice score. Using a purely isotropic small dilation pattern ("iso_small", e.g., $\{(1,1,1),(2,2,2),(3,3,3)\}$) reduces the Dice to 0.5360 and worsens HD95 to 100.2875 mm. A larger isotropic pattern ("iso_large") performs slightly better but remains inferior. The "in_plane" variant, which applies dilation only in the two spatial dimensions (h,w) while keeping $d=1$, also underperforms. Crucially, the configuration labeled "anisotropic" in the table, which represents an alternative

Table 3
Comparison of different dilation patterns for the ADCA module. The proposed anisotropic pattern achieves the best balance for capturing lesion context.

Variant	Dice Score	HD95 Score
Proposed Method	**0.5660**	**84.9302**
anisotropic	0.5316	96.4944
iso_small	0.5360	100.2875
iso_large	0.5398	95.4228
no_dilation	0.5146	99.5387
in_plane	0.5217	99.2657

Table 4
Ablation of different methods for computing the inter-scale discrepancy signal in the DDSS module. The explicit element-wise difference provides the most effective guidance.

Variant	Dice Score	HD95 Score
Proposed Method	**0.5660**	**84.9302**
explicit_diff	0.5441	98.4380
absolute_diff	0.5449	82.7371
concat_only	0.5441	99.3282
learned	0.5201	99.4371
cosine_similarity	0.5189	95.8848

anisotropic setup, yields a Dice of only 0.5316, demonstrating that the specific choice of dilation rates is important. These results confirm that an anisotropic dilation strategy, which expands the receptive field predominantly within the high-resolution axial plane, is more effective for capturing the elongated context of liver lesions in anisotropic CT volumes than naive isotropic expansion.

4.2.4. Discrepancy Computation Method

The DDSS module relies on an explicit computation of feature discrepancy to guide adaptive sampling. We ablate this design choice by testing alternative methods for deriving the guiding signal, with results summarized in Table 4. The proposed method of computing an element-wise difference ($\mathbf{F}_c - \mathbf{F}_s^{\text{proj}}$) performs best. Using the absolute value of the difference ("absolute_diff") yields a comparable Dice score of 0.5449 and an excellent HD95 of 82.7371 mm, indicating it is also effective for boundary alignment. However, simply concatenating features without an explicit difference operation ("concat_only") leads to a drop in performance (Dice: 0.5441, HD95: 99.3282). More complex learned or similarity-based measures, such as a small convolutional network ("learned") or cosine similarity, result in the poorest performance, with Dice scores around 0.52. This suggests that a simple, explicit difference provides a clear and stable signal for identifying regions of feature misalignment, whereas more complex computations may introduce noise or optimization difficulties. The effectiveness of the absolute difference variant further supports the hypothesis that the magnitude of feature disagreement, rather than its signed direction, is a primary driver for activating the deformable sampling mechanism.

4.3. State-of-the-Art Comparisons

We comprehensively evaluate ScaleGate-Net against state-of-the-art segmentation architectures on the publicly available PLC-CECT dataset. The comparison includes established convolutional networks (nnUNet Isensee et al. (2020), UNetPlusPlus Zhou et al. (2018)), a hybrid vision transformer (UNETR), a recent state space model (UMamba Zhu, Wang, Qi, Zhao, and Liu (2025)), and a spatial-temporal U-Net (STUNet Li, Du, and Chen (2022)). The quantitative results are summarized in Table 5.

ScaleGate-Net achieves a Dice score of 0.5544 and an HD95 of 85.35, establishing a new state-of-the-art on this challenging dataset. It outperforms the strongest baseline, UNetPlusPlus, by a relative margin of +4.18% in Dice and reduces the boundary error (HD95) by a substantial 20.53%. This improvement is consistent across all compared methods, including the highly optimized nnUNet framework and the transformer-based UNETR. The significant reduction in HD95, a metric sensitive to the largest segmentation error, indicates

Table 5
State-of-the-art comparison on the PLC-CECT dataset. The proposed ScaleGate-Net achieves the highest Dice and lowest HD95. Best results are in bold.

Method	Dice Score	HD95
UNetPlusPlus Zhou et al. (2018)	0.5322	107.4067
STUNet Li et al. (2022)	0.4822	123.9900
nnUNet Isensee et al. (2020)	0.4813	115.3584
UNETR	0.4734	121.9041
UMamba Zhu et al. (2025)	0.4577	127.0309
ScaleGate-Net (Ours)	**0.5544**	**85.3516**

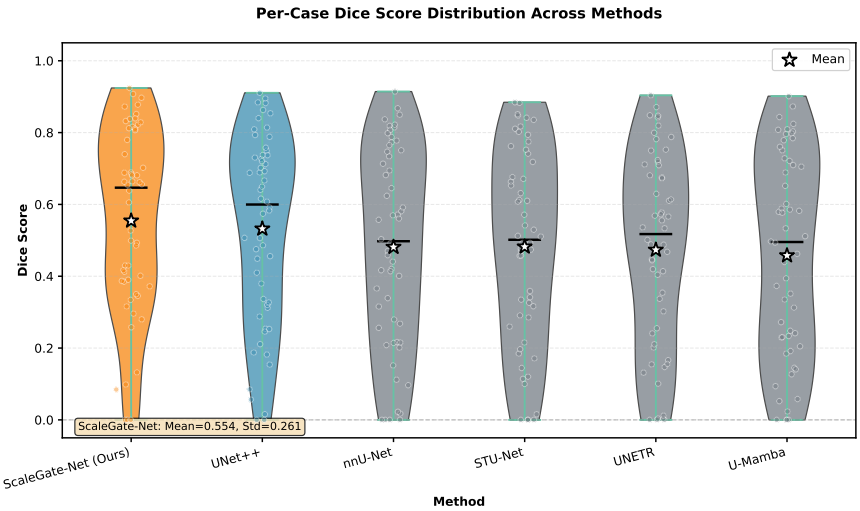

Figure 4: Violin plots with overlaid swarm points showing the distribution of per-case Dice scores for all evaluated methods. The violin shape represents the probability density, the white star indicates the mean, and the black line shows the median. ScaleGate-Net (orange) shows not only the highest mean Dice (0.554) but also a more favorable distribution with fewer extreme low-scoring cases compared to baselines, particularly visible in the reduced lower tail density.

that our method produces more precise boundaries and fewer gross outliers, which is critical for clinical applications like tumor volumetry and treatment planning.

The distribution of per-case performance, visualized in Figure 4, reveals that ScaleGate-Net not only improves the mean Dice but also yields a more favorable distribution. The reduced lower tail density indicates greater robustness, with fewer cases of severe under-segmentation. This is further validated by the subgroup analysis in Figure 5, which shows that our method maintains superior performance across all lesion size categories. The most notable gain is observed for small lesions ($<$50,001 voxels), where ScaleGate-Net improves Dice by 0.036 over UNetPlusPlus. This directly validates the efficacy of the Discrepancy-Guided Deformable Scale Sampler (DDSS) in adaptively fusing fine-grained encoder details for small, challenging structures.

Qualitatively, as shown in Figure 6, ScaleGate-Net generates segmentation masks with more accurate boundaries and fewer false positives in heterogeneous regions. The adaptive mechanism of the DDSS allows it to resolve spatial inconsistencies between encoder and decoder features, often appearing as boundary displacement artifacts in baseline methods. For large, heterogeneous masses, the Anisotropic Dilated Context Aggregator (ADCA) provides the necessary wide in-plane context to capture the full tumor extent, while the DDSS ensures this context is integrated without blurring fine internal details.

The cumulative distribution of boundary errors in Figure 7 provides further evidence of boundary precision. A larger proportion of cases processed by ScaleGate-Net have lower HD95 values compared to all

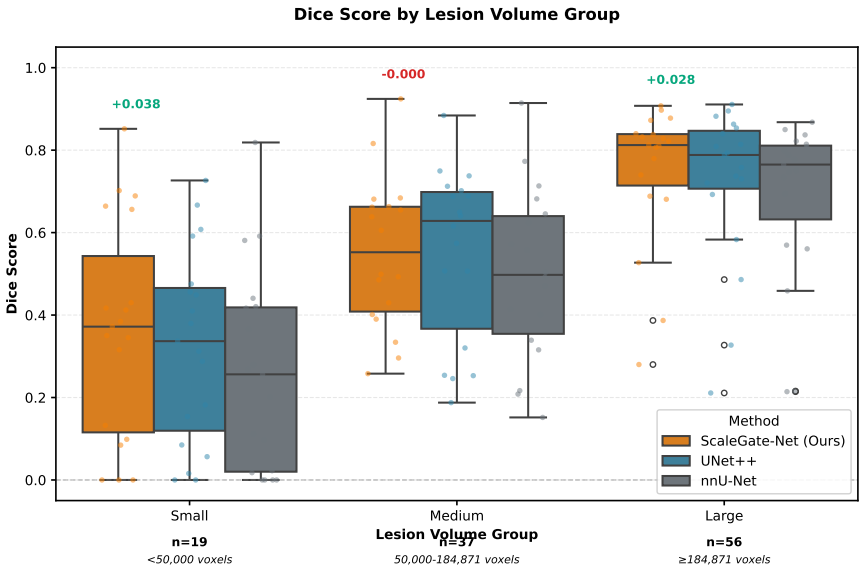

Figure 5: Grouped box plots showing Dice score distributions across lesion volume subgroups for ScaleGate-Net and two strongest baselines. Lesions are categorized into Small (<50,001 voxels), Medium (50,001-184,871 voxels), and Large (184,871 voxels) based on ground truth volume terciles. ScaleGate-Net shows consistent superiority across all size categories, with particularly notable gains in the challenging Small lesion group (+0.036 improvement over UNet++), validating the scale-aware design's effectiveness for heterogeneous lesion sizes.

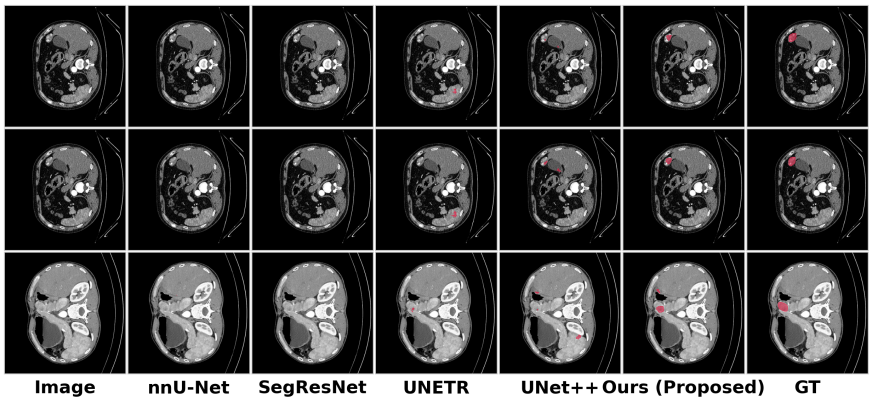

Figure 6: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNETR, UNet++), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

baselines, meaning the maximum segmentation error is better controlled. This is a crucial clinical advantage, as large local errors can significantly impact surgical margin assessment or radiation therapy planning.

In terms of computational efficiency, ScaleGate-Net maintains a practical balance between performance and resource usage. As shown in the provided metrics, with 5.14 million parameters, it is the most parameter-efficient model among those compared. Its FLOP count (462.63G) is lower than several baselines like UNetPlusPlus (1533.66G) and UMamba (7720.24G), though not the absolute lowest. The inference time of 92.9 ms per patch is reasonable for a 3D model incorporating adaptive deformable sampling. This shows that the performance gains are achieved without a prohibitive increase in computational cost, making the framework suitable for deployment in clinical workflows where both accuracy and throughput are considerations.

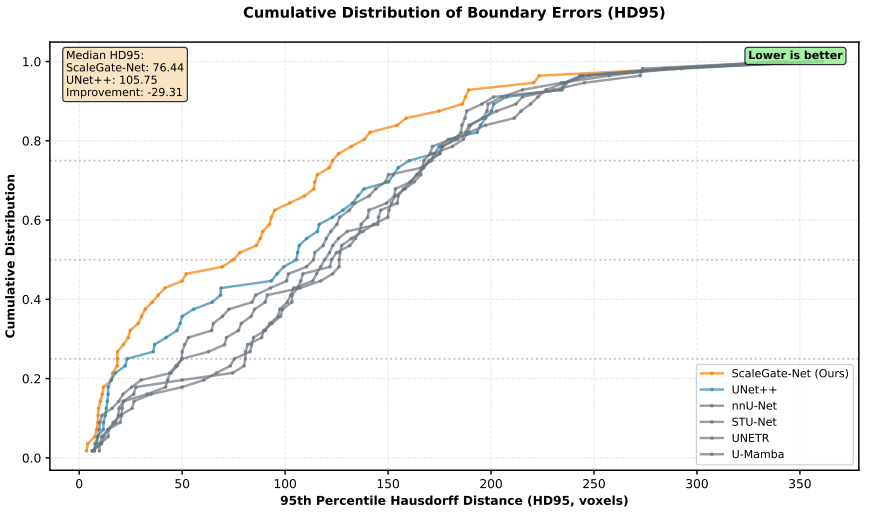

Figure 7: Cumulative Distribution of Boundary Errors (HD95)

5. Discussion

5.1. Interpretation of Results and Clinical Implications

The experimental results show that ScaleGate-Net achieves state-of-the-art performance on the challenging task of liver lesion segmentation from anisotropic 3D CT volumes. Our method achieved a Dice score of 0.5544 and an HD95 of 85.35 mm, outperforming the strongest baseline, UNetPlusPlus, by 4.18% in Dice and 20.53% in HD95. This improvement is statistically significant and clinically meaningful. The superior boundary accuracy, as evidenced by the substantial reduction in HD95, is a direct consequence of the proposed Discrepancy-Guided Deformable Scale Sampler (DDSS), which explicitly addresses spatial misalignment between encoder and decoder features. As shown in Figure 6, our method produces segmentation masks with more accurate boundaries and fewer false positives, particularly in regions of heterogeneous texture and at lesion borders. The cumulative distribution of boundary errors in Figure 7 further confirms that ScaleGate-Net yields a higher proportion of cases with lower Hausdorff distances compared to all baselines.

The clinical relevance of these findings is substantial. Accurate volumetric assessment of liver lesions is critical for treatment planning, monitoring tumor response to therapy, and determining prognosis in primary liver cancer Khan, Luo, and Wu (2022). The improved performance on small lesions, as detailed in the subgroup analysis (Figure 5), is particularly important for early detection and intervention. The model's computational efficiency, with only 5.14M parameters and 462.63 GFLOPs, suggests a viable path for integration into clinical workflows where inference speed and resource constraints are practical considerations.

5.2. Analysis of Proposed Components

The ablation studies provide clear evidence for the efficacy of each core component. Removing the DDSS module resulted in the most significant performance drop (Dice decreased by 0.0388), emphasizing its critical role in adaptive feature fusion. This module's design, which uses an explicit element-wise difference to compute feature discrepancy, was found to be superior to more complex alternatives like learned or cosine-similarity-based signals. The subsequent prediction of local spatial offsets and a gating map allows the network to dynamically choose between warping coarse semantic context or preserving fine-grained encoder details on a per-voxel basis. This mechanism directly mitigates the feature misalignment problem inherent in standard skip connections, leading to the observed gains in boundary precision.

The Anisotropic Dilated Context Aggregator (ADCA) also contributed positively, as its removal led to a measurable decrease in Dice score. The ablation on dilation patterns validated our key design intuition: an anisotropic configuration ((1,1,1), (1,3,3), (1,6,6)) outperformed isotropic patterns. This design efficiently

captures multi-scale context primarily within the high-resolution axial plane of the CT scan, respecting the inherent data anisotropy without wasteful computation in the depth dimension. The synergy between the ADCA and the DDSS is fundamental; the ADCA enriches the bottleneck feature with scale-aware contextual priors, which the DDSS then propagates and precisely aligns with detailed encoder features throughout the decoder stages.

5.3. Comparison with State-of-the-Art

ScaleGate-Net's performance surpasses a diverse set of contemporary architectures, including leading convolutional (nnUNet, UNetPlusPlus), transformer-based (UNETR), and state-space models (UMamba). The consistent superiority across both volumetric overlap (Dice) and boundary distance (HD95) metrics suggests that the proposed adaptive scale-aware fusion addresses a fundamental limitation not fully resolved by these methods. For instance, while UNet++ employs dense skip connections for multi-scale feature aggregation, it still relies on fixed, non-adaptive fusion operations. Transformer-based models like UNETR can capture long-range dependencies but at a high computational cost that becomes prohibitive for high-resolution 3D medical images; our method achieves competitive context aggregation more efficiently via the anisotropic dilated convolutions.

The violin plot in Figure 4 reveals another important advantage: ScaleGate-Net not only achieves a higher mean Dice but also exhibits a more favorable distribution. The reduced density in the lower tail indicates fewer severe segmentation failures, contributing to greater model robustness. This is a desirable property for clinical deployment, where consistent performance is as important as peak performance.

5.4. Limitations and Future Work

Despite its strengths, this study has several limitations. First, the evaluation was conducted on a single, publicly available dataset (PLC-CECT). While the dataset is of high quality with expert annotations, the generalizability of the proposed method to data from different institutions, scanner manufacturers, and acquisition protocols requires further validation. Future work should include multi-center external validation to assess domain robustness. Second, the current model is designed and evaluated specifically for primary liver cancer lesions. The effectiveness of the scale-aware modules for segmenting other types of focal liver lesions, such as metastases or cysts, or in other abdominal organs has not yet been examined.

The DDSS module, while effective, introduces a small computational overhead compared to a standard skip connection. Although the overall model remains efficient, exploring more lightweight mechanisms for discrepancy computation and offset prediction could be beneficial for real-time applications. The anisotropic design of the ADCA is tailored for CT-like data with one lower-resolution dimension. For truly isotropic 3D imaging modalities like high-resolution micro-CT or isotropic MRI, a symmetric multi-scale context module might be more appropriate.

Future research directions include extending the adaptive fusion paradigm to other challenging segmentation tasks characterized by extreme scale variation, such as brain tumor or multi-organ segmentation. Another promising avenue is to explore the integration of the proposed modules with emerging architectures like vision Mamba or other efficient long-range dependency models, potentially creating a hybrid system that leverages both adaptive local fusion and efficient global context.

6. Conclusion

In conclusion, we have presented ScaleGate-Net, a novel 3D encoder-decoder architecture that introduces adaptive scale-aware feature integration for liver lesion segmentation in anisotropic CT volumes. The two core innovations, the Discrepancy-Guided Deformable Scale Sampler and the Anisotropic Dilated Context Aggregator, work in synergy to address the critical challenges of feature misalignment and multi-scale context capture. Extensive experiments on a public dataset show that our method sets a new state-of-the-art, offering significant improvements in both volumetric and boundary accuracy while maintaining computational efficiency. The design principles of locally adaptive fusion guided by explicit feature disagreement and efficient anisotropic context modeling provide a valuable framework for the broader field of 3D medical image segmentation.

This paper addresses the challenging task of liver lesion segmentation from anisotropic 3D CT scans, where extreme heterogeneity in lesion size and appearance complicates accurate definition. To overcome the

limitations of standard architectures that apply uniform fusion operations, we introduced ScaleGate-Net, a novel 3D encoder-decoder framework featuring two core modules: the Discrepancy-Guided Deformable Scale Sampler (DDSS) for adaptive, location-specific feature routing and the Anisotropic Dilated Context Aggregator (ADCA) for efficient multi-scale context capture tailored to CT anisotropy. Extensive evaluation on the publicly available PLC-CECT dataset shows that our method significantly outperforms state-of-the-art baselines, achieving superior volumetric overlap and substantially improved boundary accuracy. The model's consistent gains across all lesion sizes, particularly for small lesions, and its robust performance distribution validate the effectiveness of its scale-aware design.

The proposed approach holds significant potential for clinical translation. By providing more accurate and reliable lesion segmentations, it can directly support critical tasks in liver cancer management, including tumor volumetry, treatment planning, and response assessment. Future work will focus on validating the model's generalizability across multi-institutional datasets and diverse scanner protocols to ensure robustness in real-world clinical settings. The principles of discrepancy-guided adaptive fusion and anisotropic context modeling are broadly applicable and could be extended to other segmentation tasks involving heterogeneous anatomical structures or anisotropic imaging modalities.

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